

Rathindra Nath KARMAKAR

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Personal Details

Date of Birth: 9 December 2001

Nationality: Indian

Education

Kyushu University

PhD in Mathematics

Fukuoka, Japan

2026/10 – Present

Kyushu University

Research Student

Fukuoka, Japan

2025/10 – 2026/09

Indian Statistical Institute

Master of Mathematics (Honours) – First Division with Distinction

Kolkata, India

2022/08 – 2024/05

- Score: 96.45%

Indian Statistical Institute

Bachelor of Mathematics (Honours) – First Division with Distinction

Bangalore, India

2019/07 – 2022/05

- Score: 96.86%

Delhi Public School, Ruby Park

Higher Secondary Education

Kolkata, India

2017/04 – 2019/05

Master's Thesis

Recent Advancements in Probabilistic Combinatorics

2024

Advisor: Prof. Soumendu Sundar Mukherjee

[Thesis report](#)

Abstract. In this project, our aim is to study how randomization can serve as a powerful technique in combinatorics and analysis. We will look at some recent spectacular applications of probabilistic techniques, including (i) the work on the sunflower conjecture by Alweiss et al. (2019), (ii) the breakthrough of the expectation threshold conjecture of Kahn and Kalai by Park and Pham (2022), (iii) the recent progress on Frankl's union-closed sets conjecture due to Gilmer (2022). We will compare the randomised approach with deterministic ones of the same or similar results, and analyse the benefits and drawbacks of each method. We will also explore applications of probabilistic techniques to other branches of mathematics.

Other Reports

Modelling Road Traffic Fatalities in India

ISI Bangalore

Advisor: Prof. Rituparna Sen

2020

- Statistical modeling coursework analyzing fatality trends using R.
- [\[Link to Project Data\]](#)

Employment History

National University of Singapore (NUS)

Singapore

Research Assistant, Department of Mathematics

2024/07 – 2025/08

- **Project:** Maximum Entropy based testing in network models.
- **Advisor:** Prof. Subhroshekhar Ghosh.
- Developed a goodness-of-fit test for network models using Jaynes' maximum entropy principle.
- Utilized free-energy methods from statistical physics, nonlinear large deviations, and graphon variational principles.
- Applied Stein's method for Poisson approximation in the sparse regime.
- **Preprint:** [arXiv:2602.20844](https://arxiv.org/abs/2602.20844)

Reading Projects & Seminars

- **Statistical Optimal Transport (2024–25):** Co-organized a NUS reading seminar with Xiaoyu Dong and Hoang-Son Tran on computational and statistical OT. [\[Seminar Page\]](#)
- **Modern Discrete Probability (2023):** Directed reading under Prof. Manjunath Krishnapur (IISc Bangalore).
- **Algebraic Topology (2022):** Fundamental Groups and Covering Spaces under Prof. Aniruddha Naolekar (ISI Bangalore).

Skills & Languages

- **Technical Stack:** Python (PyTorch, NumPy), R, $\text{\LaTeX}$.
- **Languages:**
 - **Japanese:** Intermediate – **JLPT N3 Certified** (Score: 138/180, Dec 2024).
 - **English:** Proficient – TOEFL Score: 117/120 (Sep 2023).
 - **Bengali:** Native / **Hindi:** Fluent.

Honors & Awards

- **42nd Rank** (Nationwide), ISI B.Stat/B.Math Entrance Examination (2019).
- **Sriram Scholarship** offer, Chennai Mathematical Institute (CMI) Entrance (2019).
- **11th Rank** (West Bengal), Regional Mathematical Olympiad (2017).
- **1st Place**, Bosco Xplore Science Exhibition (Distance measurement using Accelerometer).
- **Academic Excellence:** 98% in ICSE (10th, 100% Math) and 95.2% in CBSE (12th, 99% Math).